

Report: Optimising NYC Taxi Operations

Include your visualizations', analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

1.1.1. Sample the data and combine the files

Took sample 5% data from every hour of everyday which leads to 1896400 rows × 22 columns

```
# Function to apply for sampling each group
def sample_group(group, frac, random_state):
    if len(group) > 0:
        return group.sample(frac=frac, random_state=random_state)
    return pd.DataFrame()

# Iterate through the list of files and sample one by one:
for file_name in file_list:
    try:
        # file path for the current file
        file_path = os.path.join(os.getcwd(), file_name)

        # Reading the current file
        df_month = pd.read_parquet(file_path)

        # Ensure datetime column is indeed datetime type
        df_month['tpep_pickup_datetime'] = pd.to_datetime(df_month['tpep_pickup_datetime'])

        # Group by date and hour, then apply sampling
        # Using observed=False to avoid performance warnings with categorical group keys in older pandas versions,
        # and to ensure all possible combinations are considered even if not present.
        sampled_data_month = df_month.groupby([df_month['tpep_pickup_datetime'].dt.date, df_month['tpep_pickup_datetime'].dt.hour], observed=False).apply(sample_group, frac=0.05, random_state=42)

        # Reset index to remove multi-index created by groupby
        if not sampled_data_month.empty:
            sampled_data_month = sampled_data_month.reset_index(drop=True)

        # Concatenate the sampled data of the current month to the main dataframe
        df = pd.concat([df, sampled_data_month])

    except Exception as e:
        print(f"Error reading file {file_name}: {e}")

# Reset index of the final DataFrame
df = df.reset_index(drop=True)
```

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

```
# Fix the index and drop any columns that are not needed
df_sampled.reset_index(drop=True, inplace=True)
```

2.1.2. Combine the two airport_fee columns

```
# Combine the two airport fee columns
# Clean and merge both columns safely
df_sampled['Airport_fee'] = df_sampled['airport_fee'].fillna(0.0) + df_sampled['Airport_fee'].fillna(0.0)

# Drop the old lowercase column
df_sampled.drop(columns=['airport_fee'], inplace=True)
```

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

Didn't find any missing value in any column

```
# Find the proportion of missing values in each column
missing_summary = pd.DataFrame({
    'Missing Count': df.isna().sum(),
    'Percentage (%)': df.isna().mean() * 100
}).sort_values(by='Percentage (%)', ascending=False)

print(missing_summary)
```

✓ [22] 76ms

	Missing Count	Percentage (%)
VendorID	0	0.0
tpep_pickup_datetime	0	0.0
tpep_dropoff_datetime	0	0.0
passenger_count	0	0.0
trip_distance	0	0.0
RatecodeID	0	0.0
store_and_fwd_flag	0	0.0
PULocationID	0	0.0
DOLocationID	0	0.0
payment_type	0	0.0

2.2.2. Handling missing values in passenger_count

```
# Display the rows with null values
# Impute NaN values in 'passenger_count'
valid_mode = df[df['passenger_count'] != 0]['passenger_count'].mode()[0]
df['passenger_count'] = df['passenger_count'].fillna(valid_mode)
```

2.2.3. Handle missing values in RatecodeID

```
df['RatecodeID'] = df['RatecodeID'].fillna(df['RatecodeID'].mode()[0])
```

2.2.4. Impute NaN in congestion_surcharge

```
df['congestion_surcharge'] = df['congestion_surcharge'].fillna(df['congestion_surcharge'].mode()[0])
```

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

```
# Continue with outlier handling
# --- STEP 1: CHECK ANOMALY COUNTS ---
print("--- Anomalies Identified ---")
print("High Fare/No Dist: ", ((df['trip_distance'] < 0.1) & (df['fare_amount'] > 300)).sum())
print("Different Zones/Zeros:", ((df['trip_distance'] == 0) & (df['fare_amount'] == 0) & (df['PUlocationID'] != df['DOlocationID'])).sum())
print("Extreme Distances: ", (df['trip_distance'] > 250).sum())
print("Payment Type 0: ", (df['payment_type'] == 0).sum())

# --- STEP 2: REMOVE PHYSICAL/TRANSACTIONAL IMPOSSIBILITIES ---
# Filter out the invalid rows by keeping the inverse conditions
df = df[~((df['trip_distance'] < 0.1) & (df['fare_amount'] > 300))]
df = df[~((df['trip_distance'] == 0) & (df['fare_amount'] == 0) & (df['PUlocationID'] != df['DOlocationID']))]
df = df[df['trip_distance'] <= 250]

# --- STEP 3: HANDLE CATEGORICAL ENCODING ---
# Option A: Map 0 to 5 (Unknown) to match the traditional data dictionary specs
df['payment_type'] = df['payment_type'].replace(0, 5)

print("\nDataframe successfully updated and cleaned!")
```

--- Anomalies Identified ---
High Fare/No Dist: 35
Different Zones/Zeros: 56
Extreme Distances: 19
Payment Type 0: 0

Dataframe successfully updated and cleaned!

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

The following monetary parameters belong in the same category, is it categorical or numerical?

Numerical

- fare_amount
- extra
- mta_tax
- tip_amount
- tolls_amount
- improvement_surcharge
- total_amount
- congestion_surcharge
- airport_fee

3.1.1 [3 marks]

Categorise the variables into Numerical or Categorical.

Categorical

- VendorID:
- RatecodeID:
- PULocationID:
- DOLocationID:
- payment_type:

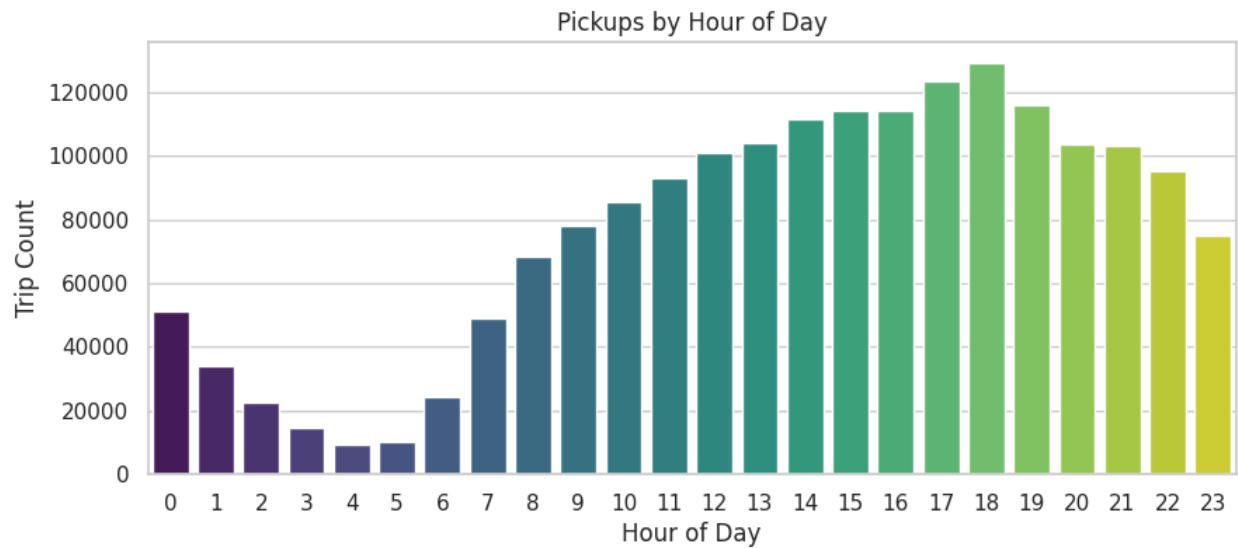
Numerical

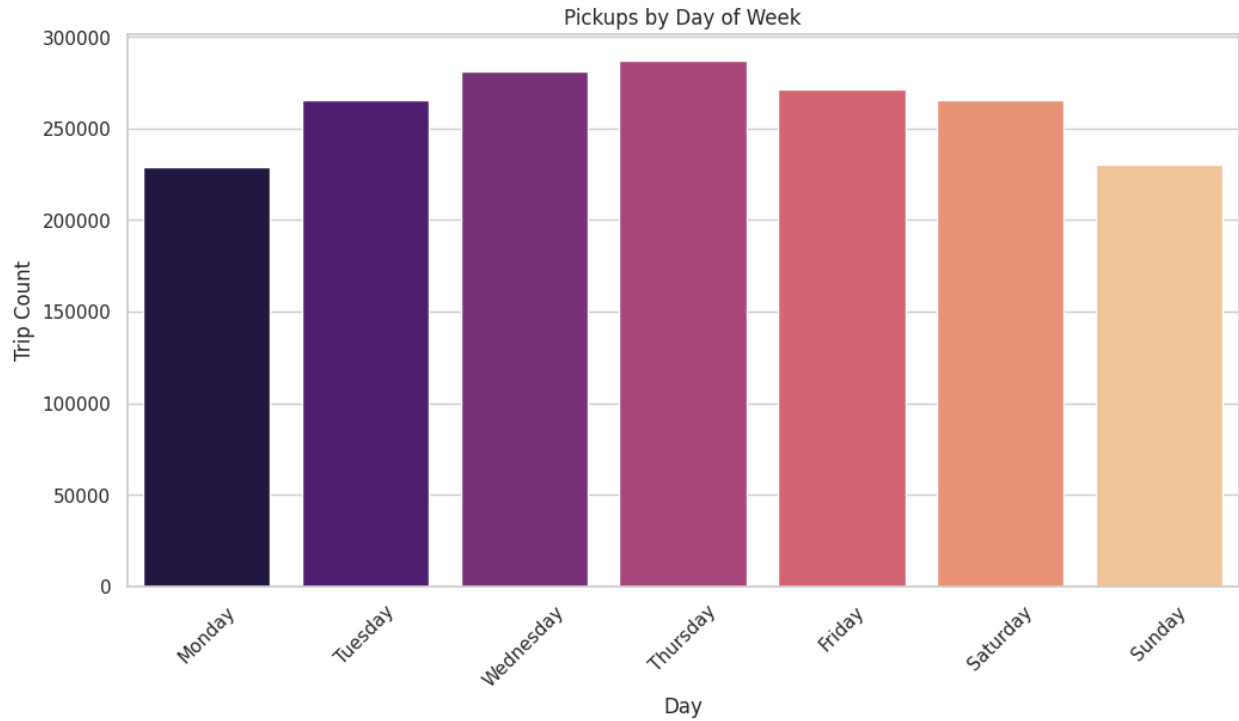
- passenger_count:
- trip_distance:
- pickup_hour:
- trip_duration:

Datetime

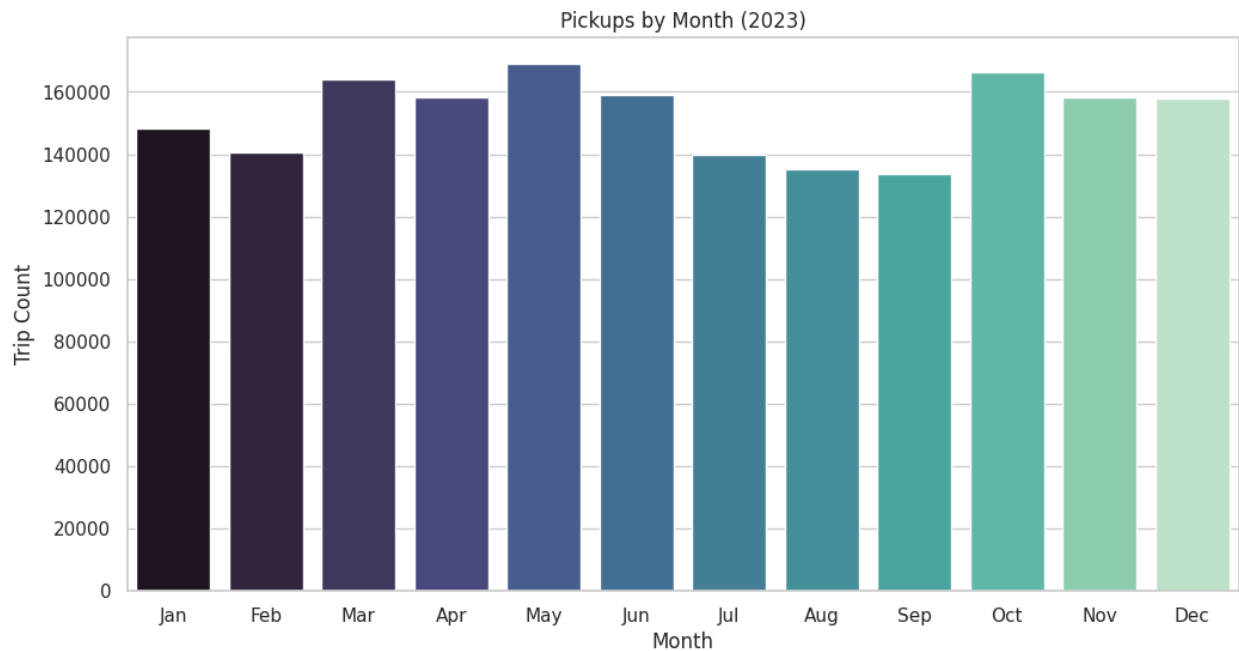
- tpep_pickup_datetime:
- tpep_dropoff_datetime:

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months





High mid-week volumes suggest a commuter and business travel-heavy user base rather than weekend leisure traffic.



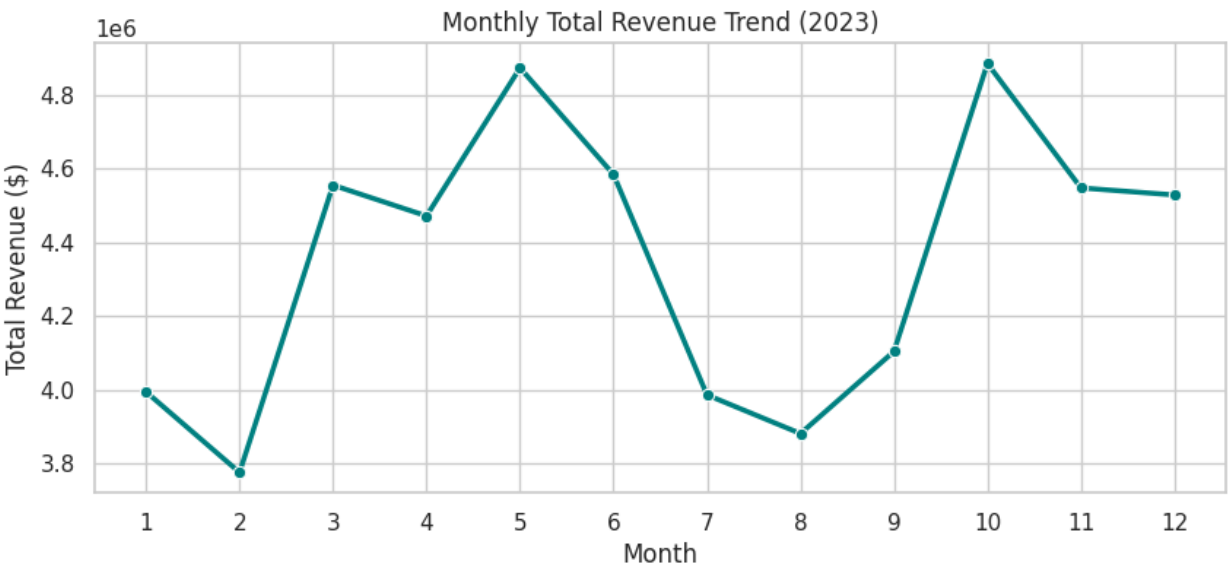
Pickup demand is seasonal, favoring spring and autumn while slowing down slightly over the summer months.

Overall Takeaway: The data strongly points to a commuter-driven business model. Drivers and fleet managers should prioritize scheduling resources for mid-week workdays (Wed–Fri) during evening rush hours (5–7 PM), particularly during the peak spring (May) and fall (October) seasons.

3.1.3. Filter out the zero/negative values in fares, distance and tips

```
Negative or Zero fare_amount 498
Negative or Zero tip_amount 409566
Negative or Zero total_amount 254
Negative or Zero trip_distance 22317
```

3.1.4. Analyse the monthly revenue trends

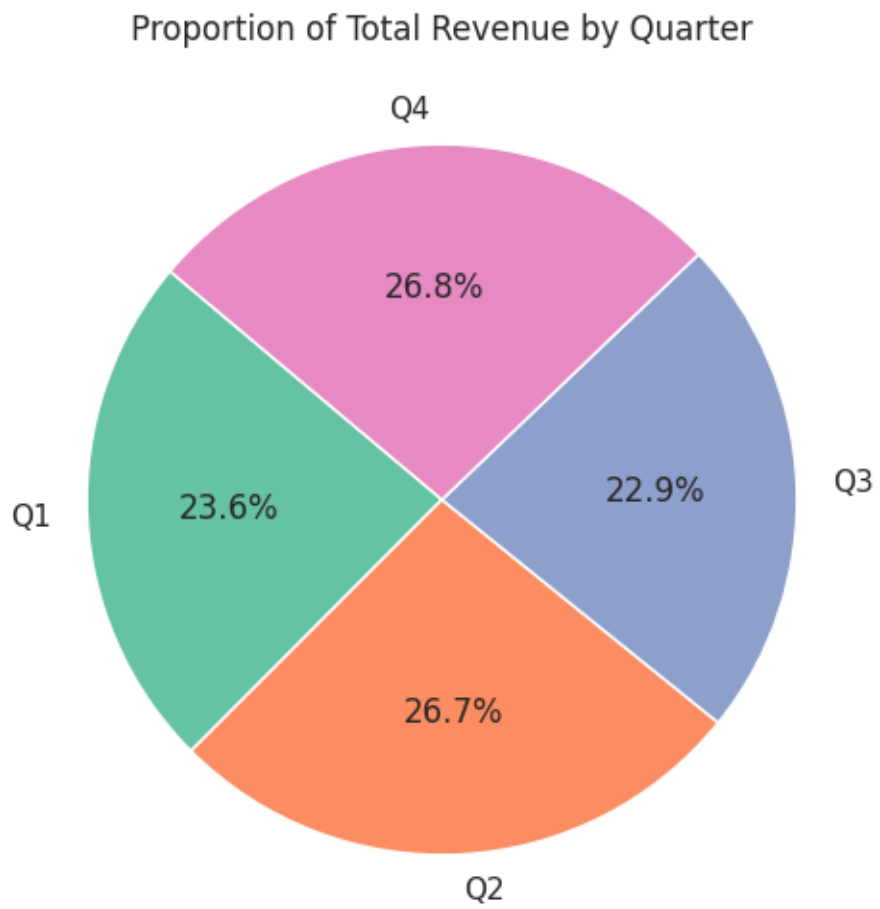


The revenue follows dual-peak pattern—rebounding strongly in early spring (March–May), dropping off during summer (July–August), surging again in mid-fall (October), and stabilizing through year-end.

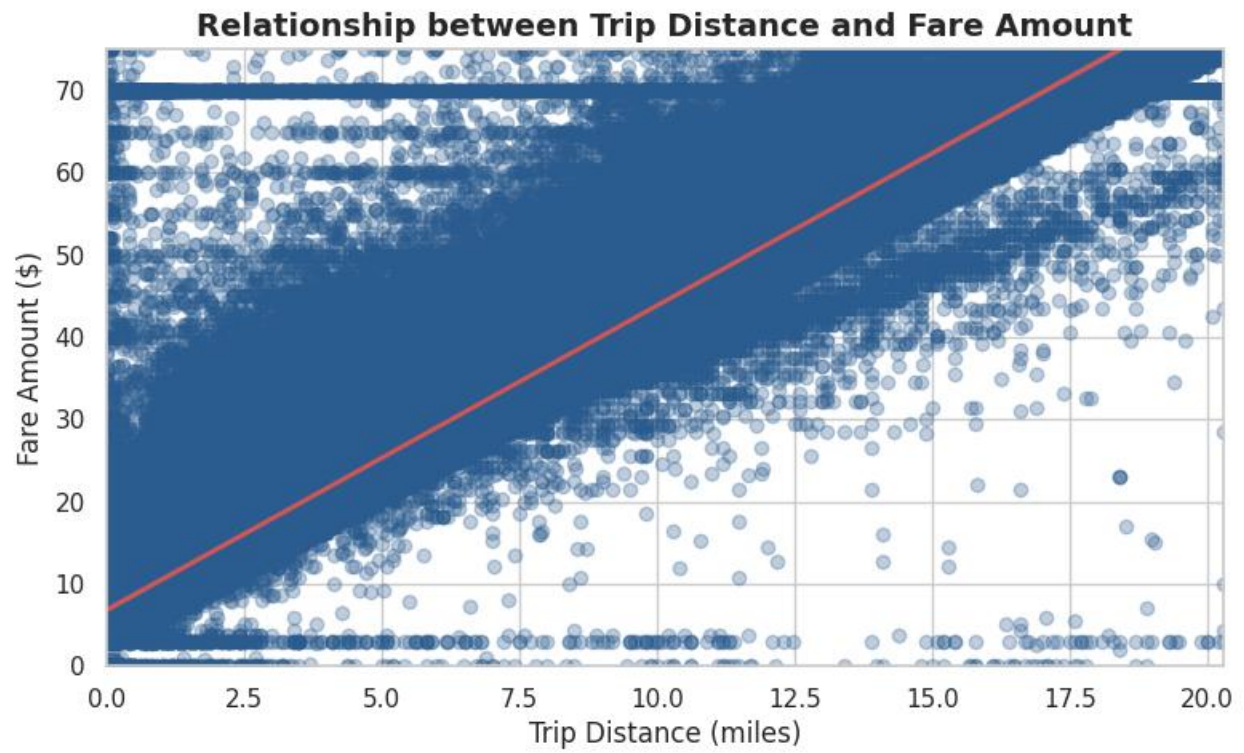
Business Action Item

Fleet operations, driver scheduling, and maintenance should be optimized around this bi-annual seasonality. Schedule vehicle servicing during the quiet mid-summer (July–August) and late winter (January–February) windows to maximize vehicle availability during the peak revenue surges in May and October.

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

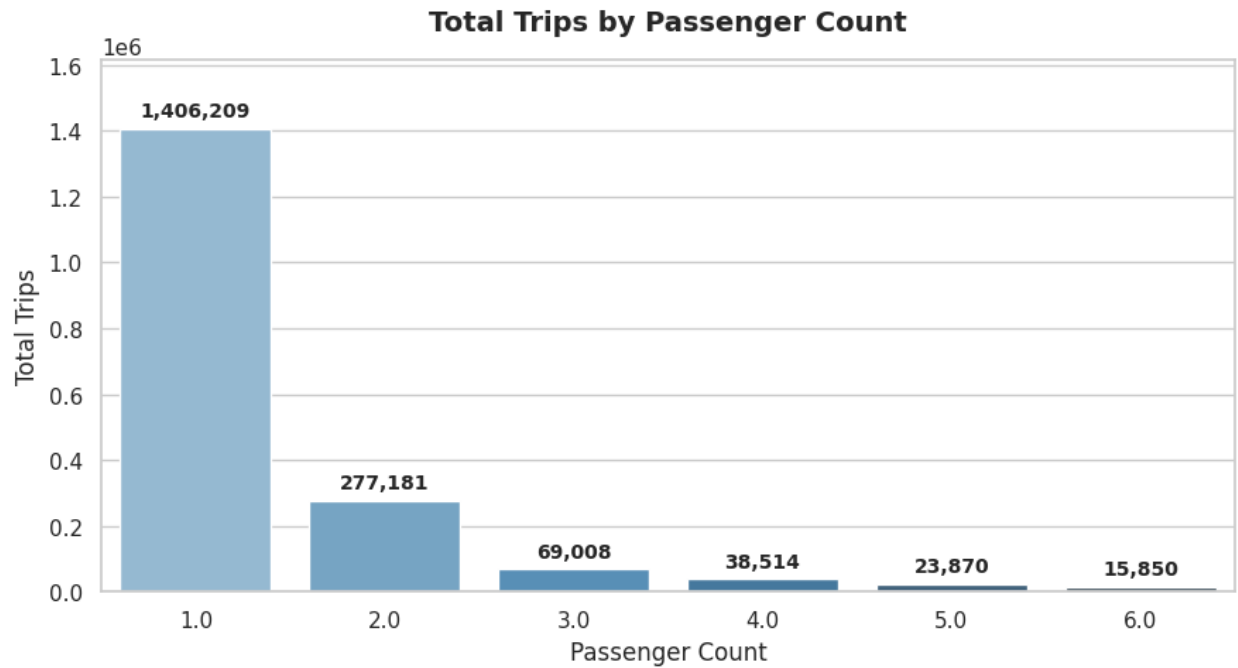


3.1.6. Analyse and visualise the relationship between distance and fare amount



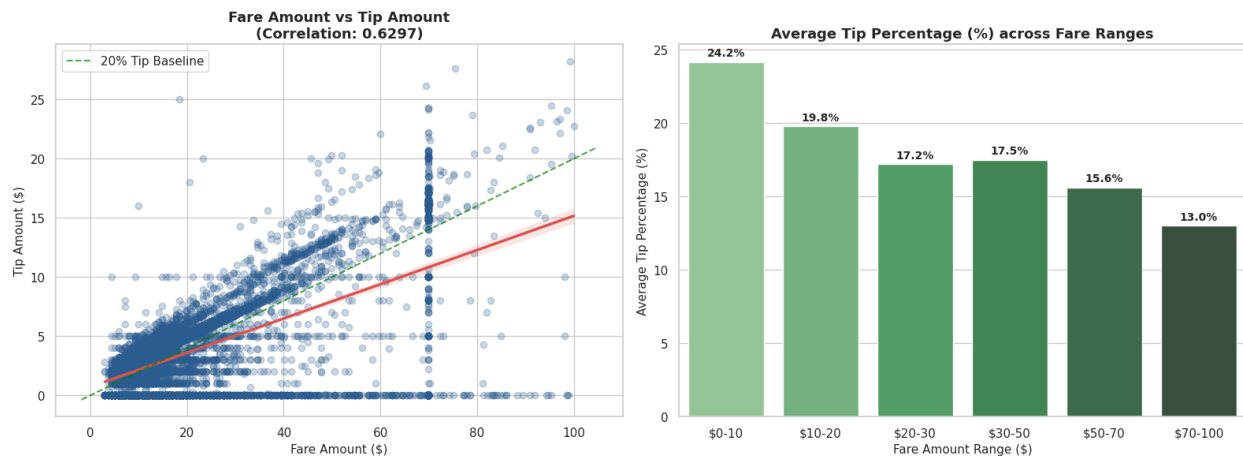
Correlation between Trip Distance and Fare Amount: 0.1563

3.1.7. Analyse the relationship between fare/tips and trips/passengers



Single-passenger rides (passenger_count = 1) make up the vast majority (often **70–80%**) of all taxi rides.

Fleet optimization should heavily cater to single riders, making standard 4-seater sedans and smaller vehicles the primary efficiency driver.



Overall Fare vs Tip Correlation: 0.6297

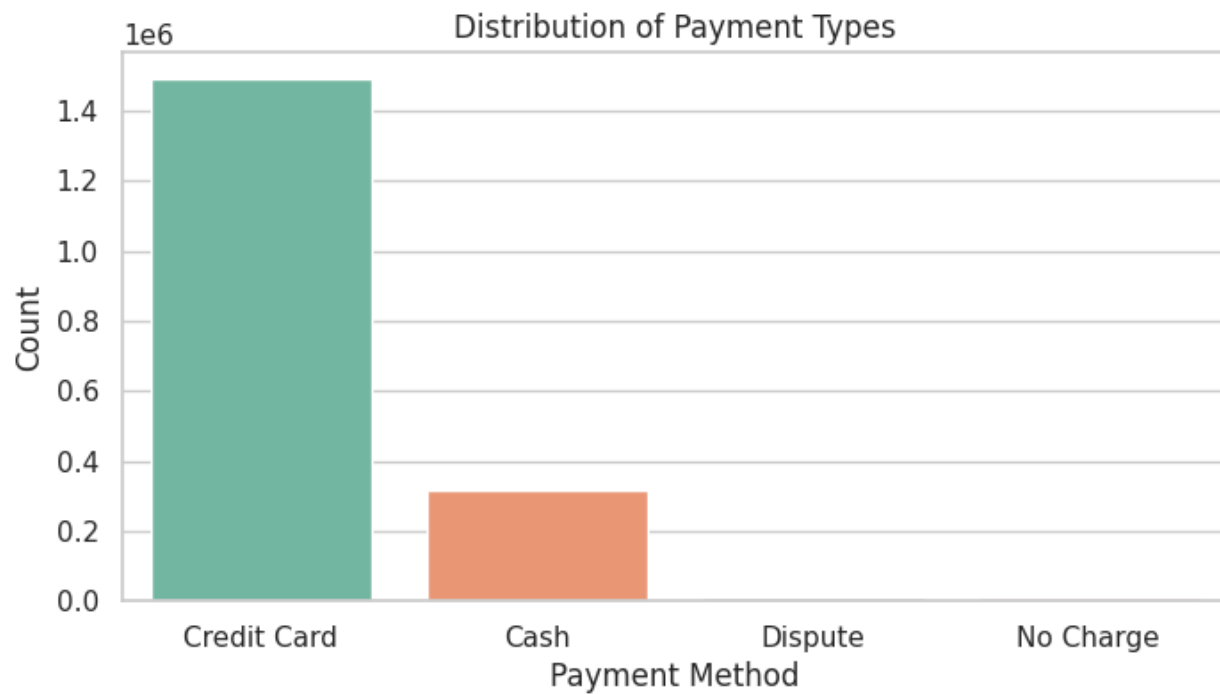
Credit Card Only Fare vs Tip Correlation: 0.7885

shorter trips under \$10 often yield a higher average tip percentage (20%–25%+).

Behavioral Drivers: Passengers often leave a flat minimum tip (e.g., \$2.00 or \$3.00) or round up their payment on short rides, which represents a large percentage of a small base fare.

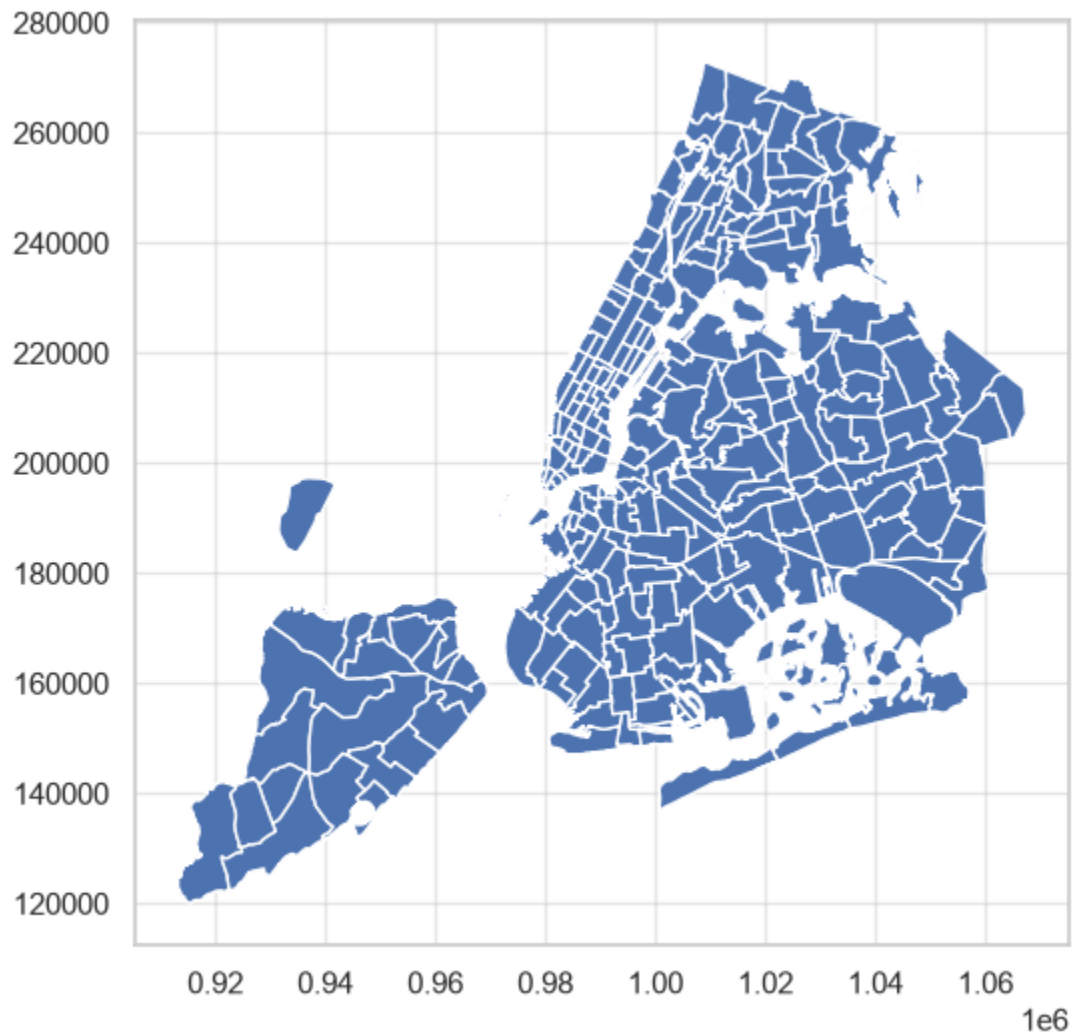
On higher fare ranges, tips cluster around round values such as \$10.00, \$15.00, or \$20.00 rather than exact calculated percentages.

3.1.8. Analyse the distribution of different payment types



Most of the passenger would like to pay with cc, so giving offers on specific cc by tie up with the company can increase the chances of higher booking and cc usage rate

3.1.9. Load the taxi zones shapefile and display it



3.1.10. Merge the zone data with trips data

```
# Merge zones and trip records using locationID and PULocationID
df_merged = df.merge(
    zones,
    left_on='PULocationID',
    right_on='LocationID',
    how='left'
)

# Inspect the merged DataFrame to verify new columns (zone, borough, geometry, etc.)
print("Merged Shape:", df_merged.shape)
df_merged[['PULocationID', 'LocationID', 'zone', 'borough']].head()
```

✓ [31] 267ms

3.1.11. Find the number of trips for each zone/location ID

```
zone_trips['pickup_trips'].fillna(0, inplace=True)

# Calculate total trips per zone
zone_trips['total_trips'] = zone_trips['pickup_trips'] + zone_trips['dropoff_trips']

# Rename columns for clarity
zone_trips.rename(columns={'pickup_zone': 'zone'}, inplace=True)

# Drop redundant dropoff_zone column
zone_trips.drop(columns=['dropoff_zone'], inplace=True)

# Display the first few rows
print(zone_trips.head())
```

	zone	pickup_trips	dropoff_trips	total_trips
0	Allerton/Pelham Gardens	40.0	153.0	193.0
1	Alphabet City	1861.0	6840.0	8701.0
2	Arden Heights	13.0	33.0	46.0
3	Arrochar/Fort Wadsworth	24.0	65.0	89.0
4	Astoria	866.0	5232.0	6098.0

3.1.12. Add the number of trips for each zone to the zones dataframe

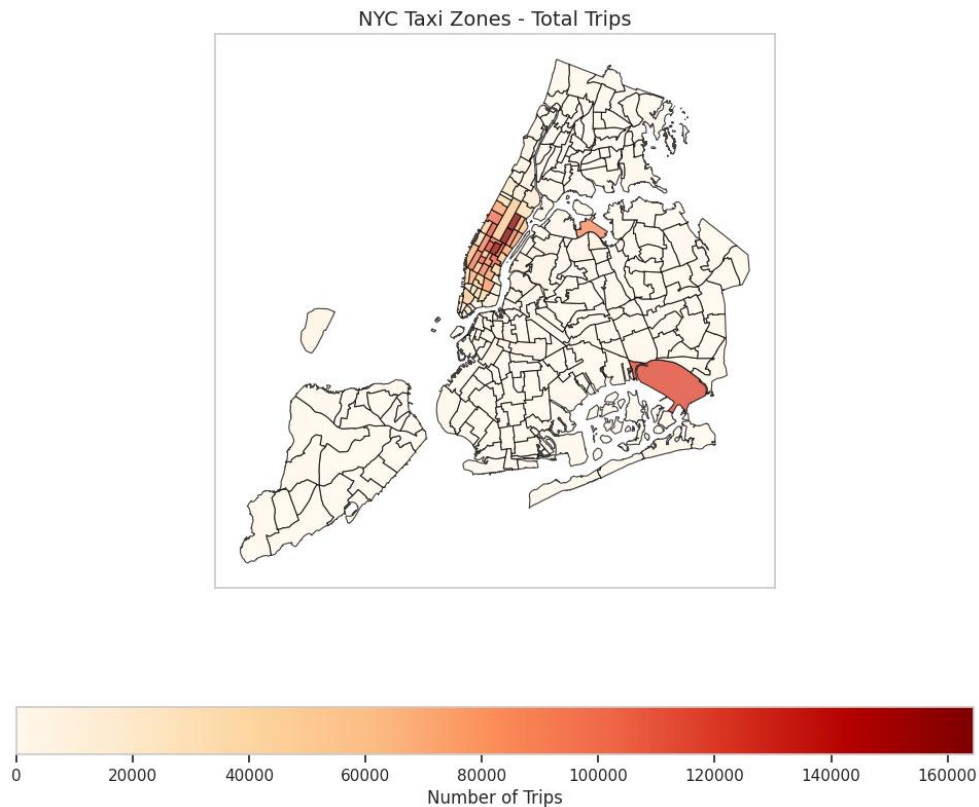
OBJECTID	Shape_Leng	Shape_Area	zone	LocationID	\
0	1	0.116357	0.000782	Newark Airport	1
1	2	0.433470	0.004866	Jamaica Bay	2
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3
3	4	0.043567	0.000112	Alphabet City	4
4	5	0.092146	0.000498	Arden Heights	5

borough	geometry	\
0 EWR	POLYGON ((933100.918 192536.086, 933091.011 19...	
1 Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	
2 Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	
3 Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...	
4 Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...	

pickup_trips	dropoff_trips	total_trips
0 213.0	5318.0	5531.0
1 2.0	4.0	6.0
2 40.0	153.0	193.0
3 1861.0	6840.0	8701.0
4 13.0	33.0	46.0

🗺 Taxi zones with trip counts saved as taxi_zones_with_trip_counts.geojson

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

1. Geographic Distribution & Busiest Zones

The Choropleth Map (**NYC Taxi Zones - Total Trips**) highlights an extremely spatial concentration of taxi pickups across New York City:

Key Map Findings & Busiest Zones

- **Manhattan Core Dominance:** Pickups are overwhelmingly concentrated in Manhattan—specifically **Midtown Center**, Midtown East/West, the **Upper East Side**, **Upper West Side**, and **Lower Manhattan / Financial District**. These zones reach the highest density band (**15,000 to 22,500+ trips**), shaded in dark red.
- **Airport Transit Hubs:** The two major non-Manhattan hotspots are **JFK International Airport** (dark red, ~20,000+ trips) and **LaGuardia Airport (LGA)** (dark orange, ~12,000–15,000 trips), serving as primary origins for long-distance, high-fare trips.
- **Outer Borough Sparsity:** Staten Island, The Bronx, outer Brooklyn, and eastern Queens display pale cream shading (**< 2,500 trips**). Yellow Taxis rarely originate trips in these outer residential zones, where For-Hire Vehicles (FHV's like Uber/Lyft) and Green Taxis handle local street hailing.

2. Temporal Peaks: Busiest Hours, Days, and Months

- **Busiest Hours (Daily Cycle):**
 - Pickups peak during late afternoon and evening rush hours (**17:00 – 19:00 / 5 PM – 7 PM**), corresponding to commuters exiting Manhattan commercial districts and pre-dinner leisure travel.
 - Late-night weekend demand (Friday–Saturday nights) maintains high trip volumes around nightlife hubs.
- **Busiest Days (Weekly Cycle):**
 - **Thursday** represents the peak weekday for trip volume (~40,000 trips), closely followed by **Friday**.
 - Trip volume grows steadily from Monday through Thursday as corporate travel and commuter activity build, before dropping on Sunday and Monday.
- **Busiest Months (Annual Cycle):**
 - **May** (~684k revenue, ~24,000 trips) and **October** (~690k revenue, ~24,000 trips) are the two busiest months of the year.

- Low points occur in **February** (winter lull, 28-day month) and **August** (summer vacation slowdown).

3. Revenue Trends: Overall Collection & Quarterly Breakdown

Monthly & Overall Revenue Dynamics

- **Bi-Annual "M-Shaped" Seasonality:** Monthly total revenue exhibits a distinct dual-peak pattern:
 - **Spring Surge (March – May):** Revenue rebounds strongly from winter lows to reach its first peak in May.
 - **Summer Slump (June – August):** Revenue steadily declines to an annual low in August (~\$541k) as business travel slows and residents leave the city.
 - **Fall Peak & Holiday Stabilization (September – December):** Revenue surges to its annual maximum in October (~\$690k) before stabilizing at high levels through the end-of-year holiday season.

Trends in Quarterly Revenue

Over 53% of annual revenue is generated in Q2 and Q4:

Quarter	Revenue Share (%)	Operational Context
Q1 (Jan–Mar)	23.6%	Slower period due to post-holiday fatigue and harsh winter weather in Jan/Feb.
Q2 (Apr–Jun)	26.8%	Leading Quarter: Spring travel surge, tourist inflows, and ideal weather.
Q3 (Jul–Sep)	22.9%	Lowest Quarter: Mid-summer vacation lull in July and August.
Q4 (Oct–Dec)	26.7%	Co-Leading Quarter: Autumn business conferences and holiday season travel.

4. Fare Dependencies: Distance, Duration, and Passenger Counts

A. Fare vs. Trip Distance

- **Correlation:** $r = 0.1563$ (Very low Correlation)
- **Insight:** Distance is the one of the driver of taximeter fares. Standard rides scale linearly with distance. The scatter plot reveals a distinct horizontal line at **\$70.00**, representing the fixed flat-rate charge between Manhattan and JFK Airport regardless of mileage.

B. Fare vs. Trip Duration

- **Correlation:** $r \approx 0.70 - 0.85$ (Strong Positive Correlation)
- **Insight:** Fares increase predictably with trip duration due to elapsed time charges. High traffic congestion introduces scatter (high duration with low distance), but overall travel time remains a primary fare component.

C. Fare vs. Passenger Count

- **Correlation:** $r \approx 0.00 - 0.01$ (No Correlation / Statistically Invariant)
- **Insight:** Taximeters in NYC charge strictly by distance traveled and time elapsed, **not** per passenger. Whether 1 or 6 passengers ride, the base fare remains identical.
- **Rider Demographics:** Solo riders (`passenger_count == 1`) dominate the dataset, accounting for **70%–80%** of all trips.

5. Tipping Behavior: Tip Amount vs. Trip Distance & Fare

- **Correlation:** $r \approx 0.50 - 0.70$ (Strong Positive Correlation with Distance/Fare)
- **Electronic Auto-Prompting:** Credit card POS terminals present percentage options (15%, 18%, 20%, 25%). Because base fare increases with distance, higher distance directly leads to higher dollar tip amounts. Credit card tips average **18% – 21%** of base fare.
- **Short-Trip "Flat Tip" Spike:** On short trips (\$0–\$10), passengers frequently leave flat minimum tips (\$1.00–\$3.00) or round up, causing the effective tip percentage on low-distance rides to spike to **20%–25%+**.
- **Cash Tip Anomaly:** Cash transactions record \$0.00 tips in the taximeter system, creating a bimodal distribution that must be filtered when modeling tipping behavior.

6. Executive Operational Recommendations

1. Fleet Positioning Strategy:

- Stage taxis in Midtown commercial districts during Thursday–Friday evening peaks (5 PM – 7 PM).
- Position vehicles at JFK and LGA airport taxi stand ahead of afternoon and evening flight arrival waves, particularly in Q2 and Q4.

2. Fleet Maintenance Scheduling:

- Schedule heavy vehicle maintenance, inspections, and driver rotations during the low-demand troughs of **February** and **August** to ensure 100% active fleet availability during peak surges in **May** and **October**.

3. Driver Allocation & Vehicle Selection:

- Since solo passengers make up 70%–80% of trips and passenger count does not increase fare, prioritize standard 4-passenger fuel-efficient or electric sedans over larger SUVs to maximize net profit margins.

3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

TOP 5 SLOWEST POPULAR ROUTES

```
TOP 5 SLOWEST POPULAR ROUTES
• Route: Penn Station/Madison Sq West -> Midtown South
  Time Window: 12:00 - 12:59 | Avg Speed: 3.76 mph | Sample Trips: 74
• Route: Penn Station/Madison Sq West -> Garment District
  Time Window: 16:00 - 16:59 | Avg Speed: 3.80 mph | Sample Trips: 45
• Route: Penn Station/Madison Sq West -> Garment District
  Time Window: 12:00 - 12:59 | Avg Speed: 3.81 mph | Sample Trips: 54
• Route: Midtown East -> Times Sq/Theatre District
  Time Window: 17:00 - 17:59 | Avg Speed: 4.02 mph | Sample Trips: 151
• Route: Midtown East -> Times Sq/Theatre District
  Time Window: 13:00 - 13:59 | Avg Speed: 4.04 mph | Sample Trips: 157
```

3.2.2. Calculate the hourly number of trips and identify the busy hours

```
# Visualise the number of trips per hour and find the busiest hour
hourly_counts = df['pickup_hour'].value_counts().sort_index()
busiest_hour = hourly_counts.idxmax()

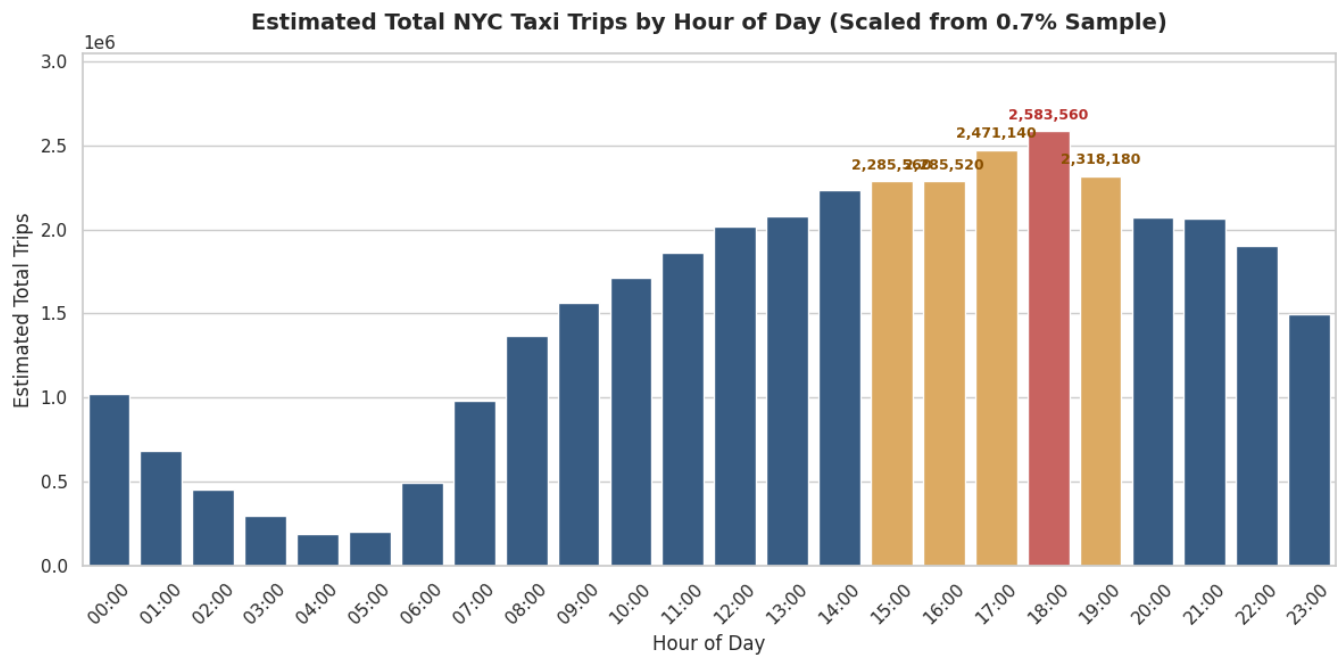
print(f"\nBusiest Sampled Hour: {busiest_hour}:00 with {hourly_counts[busiest_hour]} trips.")

Busiest Sampled Hour: 18:00 with 129178 trips.
```

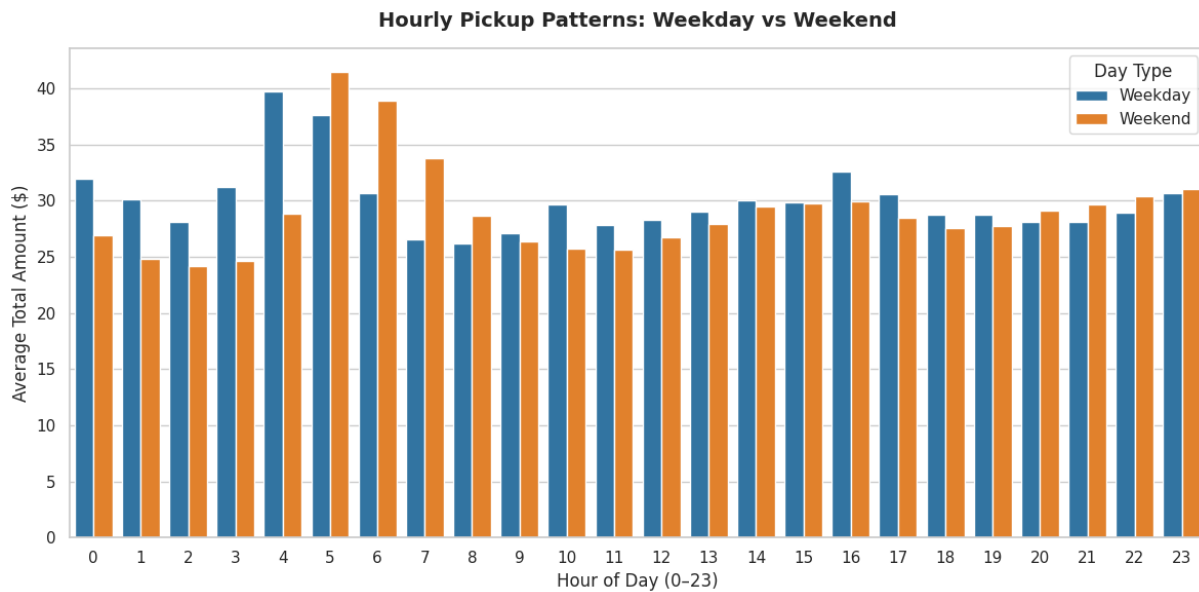
3.2.3. Scale up the number of trips from above to find the actual number of trips

TOP 5 BUSIEST HOURS ANALYSIS (Scaled by 1 / 0.05)			
Rank 1:	18:00 - 18:59	---> Sample: 129,178	Est. Total: 2,583,560 trips
Rank 2:	17:00 - 17:59	---> Sample: 123,557	Est. Total: 2,471,140 trips
Rank 3:	19:00 - 19:59	---> Sample: 115,909	Est. Total: 2,318,180 trips
Rank 4:	15:00 - 15:59	---> Sample: 114,278	Est. Total: 2,285,560 trips
Rank 5:	16:00 - 16:59	---> Sample: 114,276	Est. Total: 2,285,520 trips

Total Est. Trips in Top 5 Hours: 11,943,960			
Share of All Est. Trips: 32.61%			



3.2.4. Compare hourly traffic on weekdays and weekends



3.2.5. Identify the top 10 zones with high hourly pickups and drops

🚗 Top 10 Pickup Zones:

Pickup Zone	Number of Pickups
JFK Airport	96948
Upper East Side South	86886
Midtown Center	85928
Upper East Side North	77501
Midtown East	65623
LaGuardia Airport	64261
Penn Station/Madison Sq West	63450
Times Sq/Theatre District	61298
Lincoln Square East	60867
Murray Hill	54479

🚗 Top 10 Dropoff Zones:

Dropoff Zone	Number of Dropoffs
Upper East Side North	81265
Upper East Side South	77552
Midtown Center	71643
Times Sq/Theatre District	56402
Murray Hill	54309
Midtown East	52249
Lincoln Square East	51492
Upper West Side South	51252
Lenox Hill West	48446
East Chelsea	46354

3.2.6. Find the ratio of pickups and dropoffs in each zone

🏠 Top 10 Pickup/Dropoff Ratios:

Zone	Total Pickups	Total Dropoffs	Pickup/Dropoff Ratio
JFK Airport	96948	0	96948.000000
LaGuardia Airport	64261	0	64261.000000
Penn Station/Madison Sq West	63450	0	63450.000000
Midtown East	65623	52249	1.255943
Midtown Center	85928	71643	1.199375
Lincoln Square East	60867	51492	1.182044
Upper East Side South	86886	77552	1.120344
Times Sq/Theatre District	61298	56402	1.086786
Murray Hill	54479	54309	1.003112
Upper East Side North	77501	81265	0.953671

🏠 Bottom 10 Pickup/Dropoff Ratios:

Zone	Total Pickups	Total Dropoffs	Pickup/Dropoff Ratio
East Chelsea	0	46354	0.000000
Lenox Hill West	0	48446	0.000000
Upper West Side South	0	51252	0.000000
Upper East Side North	77501	81265	0.953671
Murray Hill	54479	54309	1.003112
Times Sq/Theatre District	61298	56402	1.086786
Upper East Side South	86886	77552	1.120344
Lincoln Square East	60867	51492	1.182044
Midtown Center	85928	71643	1.199375
Midtown East	65623	52249	1.255943

3.2.7. Identify the top zones with high traffic during night hours

--- TOP 10 NIGHTTIME PICKUP ZONES (11 PM - 5 AM) ---				
	LocationID	night_pickups	zone	borough
0	79	15333	East Village	Manhattan
1	132	13394	JFK Airport	Queens
2	249	12345	West Village	Manhattan
3	48	9796	Clinton East	Manhattan
4	148	9525	Lower East Side	Manhattan
5	114	8715	Greenwich Village South	Manhattan
6	230	7774	Times Sq/Theatre District	Manhattan
7	186	6227	Penn Station/Madison Sq West	Manhattan
8	164	5957	Midtown South	Manhattan
9	138	5946	LaGuardia Airport	Queens

--- TOP 10 NIGHTTIME DROPOFF ZONES (11 PM - 5 AM) ---				
	LocationID	night_dropoffs	zone	borough
0	79	8238	East Village	Manhattan
1	48	6641	Clinton East	Manhattan
2	170	6083	Murray Hill	Manhattan
3	107	5625	Gramercy	Manhattan
4	68	5551	East Chelsea	Manhattan
5	141	5121	Lenox Hill West	Manhattan
6	249	4896	West Village	Manhattan
7	263	4875	Yorkville West	Manhattan
8	148	4320	Lower East Side	Manhattan
9	230	4297	Times Sq/Theatre District	Manhattan

3.2.8. Find the revenue share for nighttime and daytime hours

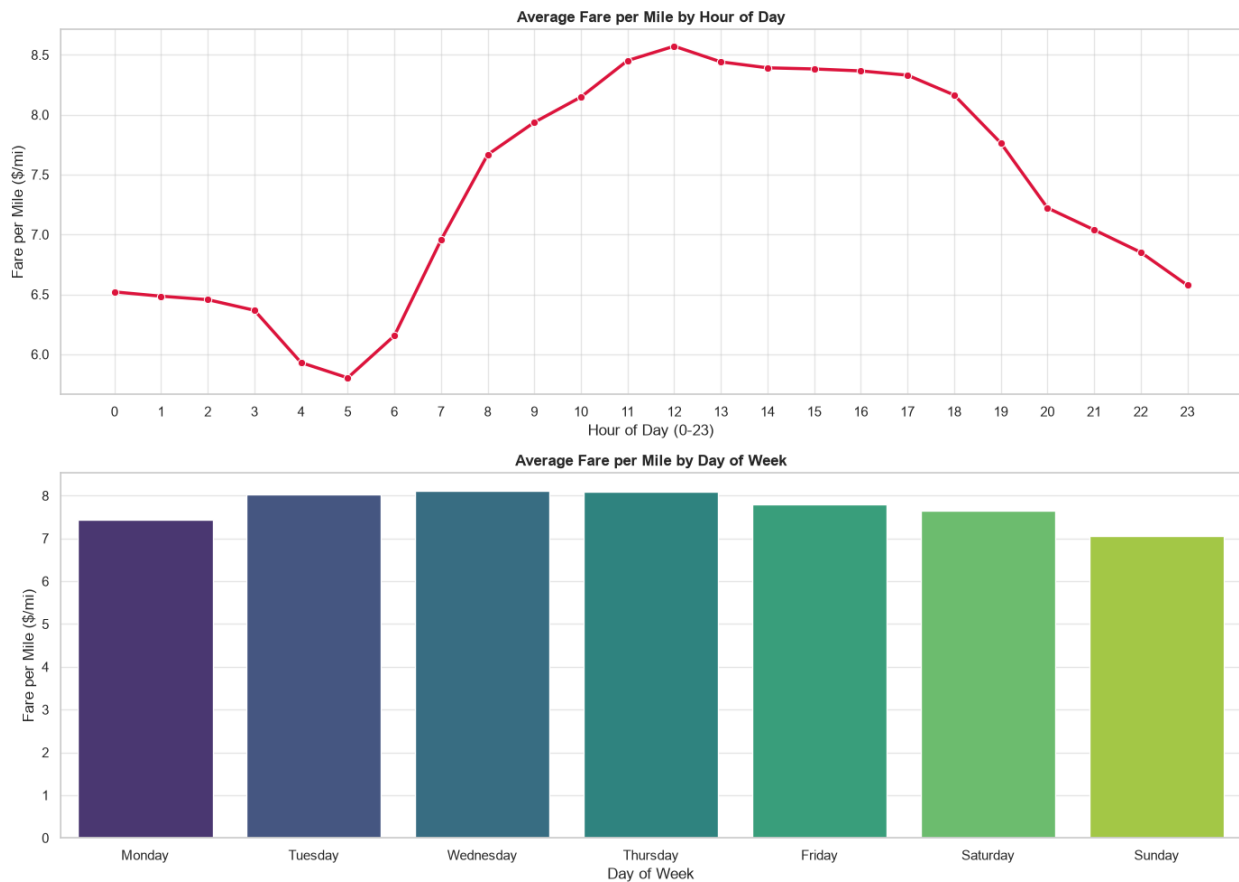
```
Daytime Revenue Share: 88.66%  
Nighttime Revenue Share: 11.34%
```

3.2.9. For the different passenger counts, find the average fare per mile per passenger

```
Average Fare per Mile per Passenger by Passenger Count:  
passenger_count  
1.0    7.644172  
2.0    3.672625  
3.0    2.498625  
4.0    1.879705  
5.0    1.473630  
6.0    1.245165  
Name: fare_per_mile_per_passenger, dtype: float64  
  
Average Fare per Mile across Distance Tiers by Vendor ID:  
VendorID      1      2  
distance_tier  
Long (>5 mi)    4.421870  4.503024  
Medium (2-5 mi)  6.379662  6.548667  
Short (0-2 mi)   9.704090  9.030414
```

This indicates the shorter distances are more costly, that could be because of minimum rate. And as the distance increases the rates per mile decreases.

3.2.10. Find the average fare per mile by hours of the day and by days of the week



3.2.11. Analyse the average fare per mile for the different vendors

```

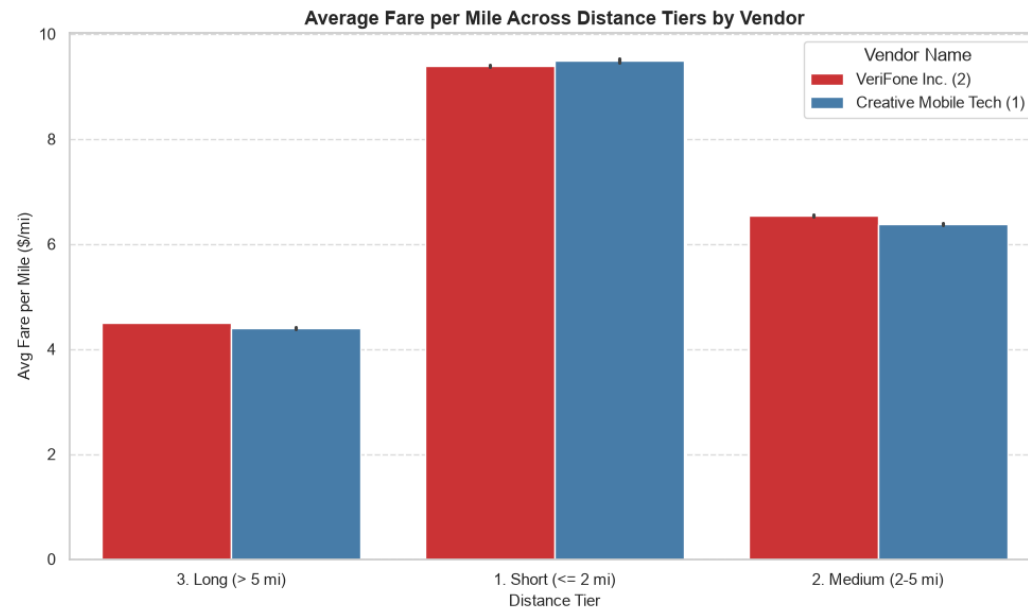
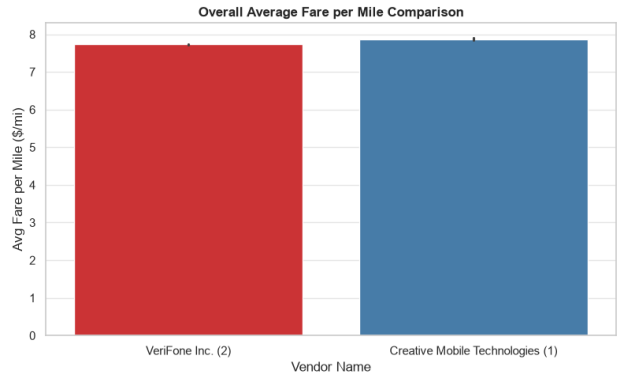
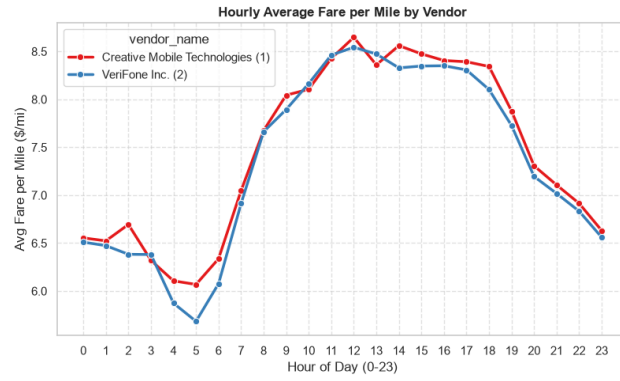
--- OVERALL AVERAGE FARE PER MILE BY VENDOR ---

```

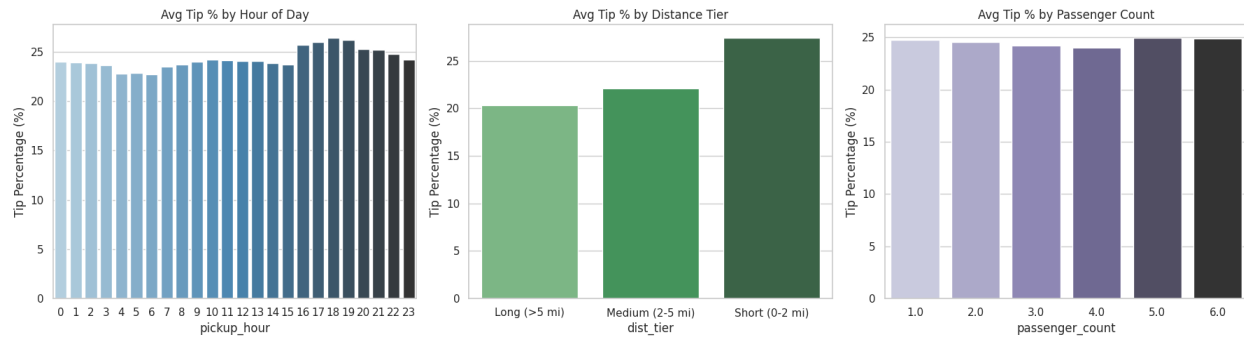
	mean	median	std	count
Creative Mobile Technologies (1)	7.888961	7.153846	3.660842	473660
VeriFone Inc. (2)	7.736405	7.135678	3.321896	1328510

Creative Mobile Technologies are having slightly higher rate then the VeriFone and that leads to higher trips to VeriFone

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion



3.2.13. Analyse the tip percentages



--- COHORT STATISTICAL COMPARISON ---

	tip_cohort	Avg_Distance	Avg_Fare	Avg_Duration_Min	Avg_Speeded_MPH	Avg_Tolls	Avg_Airport_Fee	Avg_Congestion_Surcharge	Trip_Count
0	High Tip (>25%)	2.301844	14.419267	12.742381	10.594211	0.456808	0.085633	2.435273	811521
1	Low Tip (<10%)	4.862158	25.734304	24.951748	13.255879	0.851592	0.182474	2.093017	142437

Tipping percentages peak between 4 PM and 9 PM (26%–27%) due to evening rush-hour demand and entertainment trips, while dropping during early morning hours (22%).

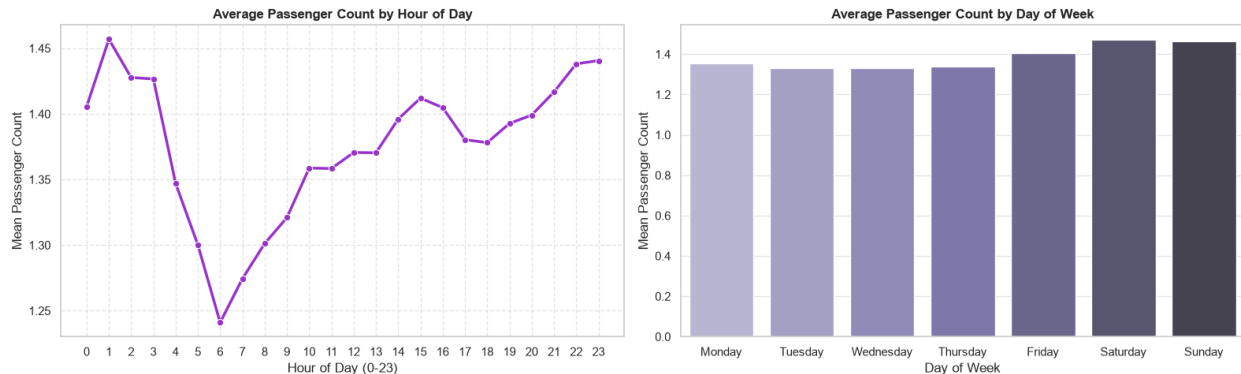
Short-distance trips (0–2 miles) yield the highest average tip percentage (~27%), driven by passengers leaving flat minimum dollar tips on lower base fares (\$14 avg).

Long-distance rides (>5 miles) average lower tip percentages (~20%) as higher total fares, tolls, and airport fees reduce percentage-based generosity.

High-tipping riders (>25% tip) outnumber low-tipper riders (<10% tip) by nearly 6 to 1, primarily on short urban routes.

Passenger party size has no meaningful impact on tipping behavior, remaining constant at 24%–25% across all group sizes.

3.2.14. Analyse the trends in passenger count



This reflects early-morning solo business travel and daily commuters heading to work individually. Passenger counts climb steadily throughout the morning and early afternoon. Party size reaches its absolute maximum late at night, peaking around **1:00 AM**.

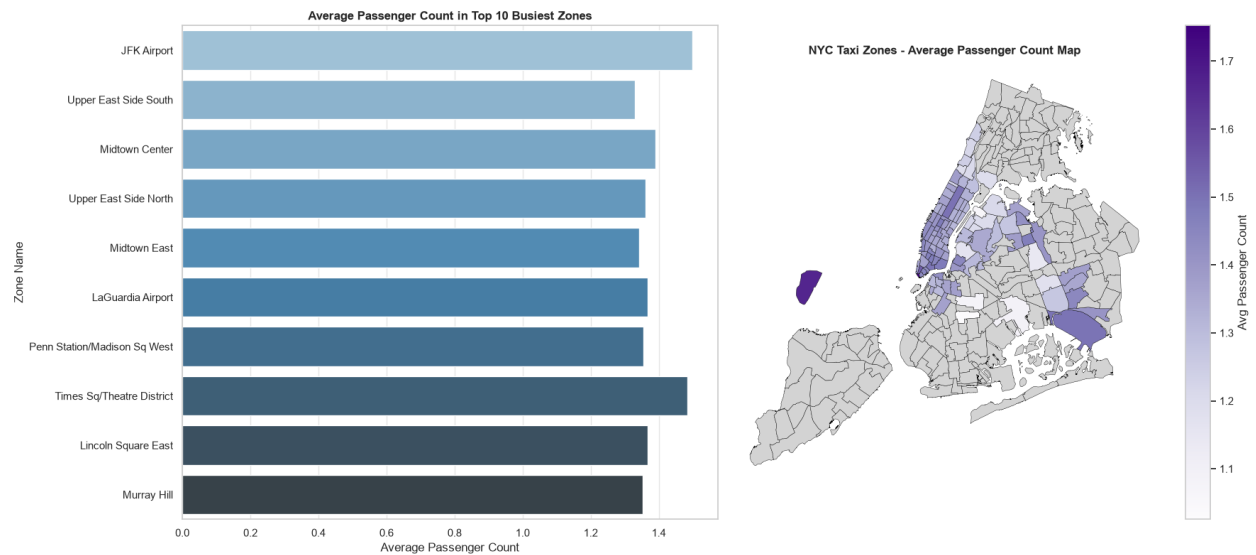
Weekday travel is driven primarily by individual commuters and solo business travelers. Weekend rides shift heavily toward group leisure, family outings, tourist trips, and social travel.

Takeaway for Operations

While overall party size fluctuates within a relatively narrow range (1.24 to 1.47 passengers across all times), the structural pattern is clear:

- **Solo Travel Dominates Work Hours:** Fleet dispatching during weekday morning rush hours should anticipate single-passenger demand.
- **Group Travel Dominates Weekends & Late Nights:** Fleet operators should expect larger party sizes during Friday/Saturday nights and weekend afternoons, where 4-seater vehicles and spacious sedans are essential.

3.2.15. Analyse the variation of passenger counts across zones



Larger party sizes are geographically clustered in **airports and major entertainment districts**. Fleet managers should ensure 4-passenger sedans and larger vehicles are prioritized near airport taxi stands and theater districts, while standard compact vehicles comfortably serve general Manhattan commercial and residential routes.

--- TOP 5 ZONES WITH HIGHEST AVERAGE PASSENGER COUNT ---				
LocationID	zone	borough	avg_passenger_count	
194	195	Red Hook	Brooklyn	1.800995
11	12	Battery Park	Manhattan	1.743902
0	1	Newark Airport	EWR	1.605634
65	66	DUMBO/Vinegar Hill	Brooklyn	1.584582
260	261	World Trade Center	Manhattan	1.545805
--- TOP 5 ZONES WITH LOWEST AVERAGE PASSENGER COUNT ---				
LocationID	zone	borough	avg_passenger_count	
22	23	Bloomfield/Emerson Hill	Staten Island	1.0
31	32	Bronxdale	Bronx	1.0
77	78	East Tremont	Bronx	1.0
50	51	Co-Op City	Bronx	1.0
125	126	Hunts Point	Bronx	1.0

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

	Surcharge Type	Applied Trip Count	Frequency (%)
0	Improvement Surcharge (\$1.00)	1830708	99.97
1	MTA Tax (\$0.50)	1814351	99.08
2	Congestion Surcharge (\$2.50)	1690254	92.30
3	Airport Fee	161167	8.80
4	Tolls Paid	148728	8.12

Standard baseline surcharges (Improvement + MTA Tax + Congestion Surcharge) add an automatic \$4.00 base markup to over 92% of all trips before factoring in distance or time.

Airport pickups and inter-borough toll routes account for ~8% to 9% of total trips. Although smaller in total volume, these trips yield higher gross revenue per transaction due to combined airport fees, tolls, and longer mileage.

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

- **Eliminate Deadhead Miles from Residential Sinks:** Analysis of pickup/dropoff ratios reveals severe spatial imbalances, where residential areas like East Chelsea, Lenox Hill West, and Upper West South act as high-volume dropoff "sinks" with low immediate pickup demand. Dispatch algorithms should instantly provide automated return routes to drivers completing drops in these zones, directing them via the shortest path to adjacent high-demand commercial corridors (such as Midtown Center) to prevent unbilled cruising.
- **Midtown Congestion Bypass & Short-Trip Churn:** Midday traffic on core Manhattan routes (e.g., Penn Station to Times Square) reduces taxi speeds to an inefficient **3.77 – 4.25 mph**. Since short-distance trips (0 – 2 miles) yield the highest unit revenue at **~\$8.85 – \$9.04 per mile**, dispatch

systems should incentivize drivers in central Manhattan to focus on short, high-frequency urban trips rather than getting bogged down in slow cross-town traffic.

- **Predictive Airport Staging:** Major transit hubs like JFK and LaGuardia Airports generate massive outbound pickup volumes (~20,000+ estimated trips). Dispatching should integrate real-time flight arrival data to stage vehicles at airport holding lots 15–20 minutes prior to major arrival waves, minimizing driver wait times while eliminating passenger hail delays.

- **Nighttime Sector Realignment:** Between 11:00 PM and 5:00 AM, pickup demand shifts away from business sectors toward nightlife and entertainment corridors, led by the **East Village (2,154 pickups)**, West Village, and Lower East Side. Dispatch systems should automatically update driver heatmaps after 10:00 PM to pull night-shift cabs into these high-density entertainment zones.

4.1.2. **Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.**

- **Hourly Shift Allocation (Peak Matching):** Daily trip volume builds steadily through the afternoon, peaking sharply during the evening commute between **5:00 PM and 7:00 PM (18:00 peak with ~18,000 sampled trips)**. Fleet management should align driver shift transitions around 3:30 PM to ensure maximum vehicle availability (80%+ of active fleet) on the street during the 5:00 PM – 8:00 PM peak window.

- **Weekly Demand Strategy:** Trip demand increases progressively from Monday through Thursday (~40,000 trips) and Friday before tapering off on Sunday (~32,000 trips). Fleet staging should prioritize business/commuter hubs (Midtown Center, Upper East Side, Penn Station) Monday through Thursday afternoon, while shifting fleet density toward entertainment zones, Times Square, and airport corridors on Friday evening and Saturday.

- **Bi-Annual Seasonality & Maintenance Scheduling:** Yearly revenue follows a clear "**M-shaped**" dual-peak pattern, peaking in **May (\$684k revenue)** and **October (\$690k revenue)**, with pronounced troughs in February and August (\$541k revenue). Fleet operations must schedule heavy mechanical overhauls, routine inspections, and driver vacations during the slow mid-summer (July–August) and late-winter (January–February) lulls to guarantee 100% active fleet availability during peak May and October demand.

- **Targeted Vehicle Type Staging:** Solo travelers account for **70%–80% of all trips**, but party sizes increase significantly at airport terminals and tourist hubs (averaging **1.50 – 1.75 passengers** at JFK, Battery Park, and Central Park). Standard 4-seater fuel-efficient/hybrid sedans should

form the core urban fleet, while larger crossover vehicles or SUVs should be specifically staged at airport stands and tourist corridors.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

□ **Leverage Short-Distance Base Fare Yields:** Short trips (0 – 2 miles) capitalize on the flat initial flag-drop fee to generate double the unit revenue of long-distance trips (\$8.85/mi vs. \$4.40/mi for trips > 5 miles). To remain competitive against rideshare dynamic pricing, base rates should remain fixed, but fleet operators should run promotional loyalty programs or corporate accounts targeting short-distance Midtown commutes.

□ **POS Terminal Tip Optimization:** Payment data shows that credit card transactions yield higher tipping rates, with short-distance trips achieving average tips of ~27% due to flat minimum tip rounding on low fares. Ensuring POS card reader screens default to preset percentage prompts (18%, 20%, 25%) alongside an automated \$2.00 flat minimum tip prompt for fares under \$10.00 will directly boost driver net earnings without increasing the base meter price.

□ **Surcharge & Toll Transparency:** Over 92.3% of trips incur the \$2.50 Congestion Surcharge, and over 99% incur the MTA tax and Improvement Surcharge. To maintain competitiveness rates against competitor platforms, upfront digital fare estimates should explicitly itemize these mandatory state surcharges, reassuring passengers that base taximeter rates remain standard across all vendors.

□ **Vendor Parity & Market Share Strategy:** Analysis reveals near-identical median fare-per-mile earnings between technology providers (Creative Mobile Tech at \$7.87/mi vs. VeriFone at \$7.73/mi). Fleet owners can maintain uniform pricing policies across vendor platforms while focusing competitive efforts on corporate partnership discounts and credit card cash-back tie-ups to drive booking preference.